Revisiting Local and Global Descriptor-Based Metric Network for Few-Shot SAR Target Classification

Jia Zheng, Ming Li, *Member, IEEE*, Xiang Li, Peng Zhang, *Member, IEEE*, and Yan Wu, *Member, IEEE*

***Abstract*— Convolutional neural network (CNN) still suffers from overfitting problems caused by limited samples in SAR target classification. Few-shot learning (FSL) aims to learn a classifier to classify images when only a few training examples are available for each class. Recent works on FSL demonstrate that using local descriptor representations can lead to more informative representations than using image-level representations. However, local descriptors typically capture only local information in an image, while disregarding global contextual information. Therefore, relying solely on local descriptors may not fully represent the features of an image, thus affecting the classification performance of the synthetic aperture radar (SAR) targets. To address this problem, we propose a novel global and local descriptor-based metric network (GLMnet) for few-shot SAR target classification. The proposed GLMnet consists of global-level metric relationships and local-level metric relationships. The global-level metric relationships are computed by the Brownian distance covariance metric module to capture the relationships between image blocks. On the other hand, the local-level metric relationships are generated by the image-to-class (I2C) metric module. To integrate these two metric relationships, we design an adaptive metric fusion module (AMFM) to achieve a more discriminative measurement performance. Furthermore, to address the issue of prototypes deviating from the true distribution due to the presence of outliers or noisy samples in the BDC metric process, we design a weighted average prototype computation module (WAPCM) to obtain reliable prototypes. Experimental results on three SAR datasets show that the proposed GLMnet can generate reliable prototypes and achieve comparable classification results in contrast to other FSL algorithms.**

***Index Terms*— Few-shot learning (FSL), global-level metric, local-level metric, synthetic aperture radar (SAR), target classification.**

I. Introduction

SYNTHETIC aperture radar (SAR) is widely used in fields such as battlefield reconnaissance, geological observation [1], disaster monitoring [2], and ocean observation [3] thanks to its all-weather, all-time, high-resolution, and long-range operational capabilities. By capturing the electromagnetic scattering characteristics of targets, SAR provides unique information that is crucial for target identification. However, due to the complicated characteristics of SAR, there are challenges and relatively low efficiency in SAR target recognition. Therefore, SAR automatic target recognition (ATR) has attracted extensive attention, leading to the emergence of various related methods.

Traditional SAR target classification methods [4], [5], [6], [7], [8] heavily depend on hand-designed features, limited by the experience in handling complex SAR images. Additionally, these methods exhibit weak abilities in target abstraction and generalization, making it challenging to capture high-level semantic information. Moreover, traditional approaches face difficulties in training and optimizing on large-scale datasets.

With the development of artificial intelligence technology, deep learning image processing algorithms achieve automatic extraction of high-level features from raw image data and perform classifier training simultaneously through end-to-end training. LeCun et al. [9] first propose convolutional neural network (CNN) in the early 1990s, and it has had a significant impact on the field of computer vision. It has pioneered new directions in image recognition and classification tasks, leading to numerous groundbreaking achievements. The emergence of AlexNet [10] in 2012 has a significant impact on CNN, significantly improving the accuracy of image classification. It lays the foundation for CNN to become a core technology in the field of computer vision and image processing. Since then, CNN has demonstrated outstanding feature extraction capabilities in various fields, such as semantic segmentation, object detection, and image recognition. Currently, CNN-based deep learning models [11], [12], [13], [14], [15], [16], [17], [18], [19], [20] have been widely applied in SAR target classification, demonstrating greater advantages over traditional methods. Furthermore, significant progress has been made in SAR target feature extraction and classification problems.

However, CNN-based deep learning models require a large amount of training data and computational resources. Additionally, these models heavily rely on annotated data, and poor quality or insufficient annotation data may negatively impact the performance of the model. In practical applications, acquiring SAR data is more challenging compared to optical images, and annotating SAR images is also more time-consuming and labor-intensive. Some categories may have only a few dozen or even fewer samples, which can easily lead to overfitting

Manuscript received 24 June 2023; revised 1 November 2023; accepted 2 February 2024. Date of publication 7 February 2024; date of current version 23 February 2024. This work was supported in part by the Natural Science Foundation of China under Grant 62172321, in part by the Civil Space 13 Five Years Pre-Research Project under Grant D040114, and in part by the Aeronautical Science Foundation under Grant 20200001081001. *(Corresponding author: Ming Li.)*

Jia Zheng, Ming Li, and Peng Zhang are with the National Key Laboratory of Radar Signal Processing, Xidian University, Xi'an 710071, China (e-mail: liming@xidian.edu.cn).

Xiang Li is with the Beijing Institute of Radio Measurement, Beijing 100854, China.

Yan Wu is with the Remote Sensing Image Processing and Fusion Group, School of Electronic Engineering, Xidian University, Xi'an 710071, China.

Digital Object Identifier 10.1109/TGRS.2024.3363436

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:45:04 UTC from IEEE Xplore. Restrictions apply.

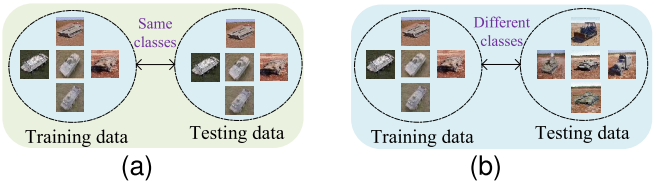


Fig. 1. Difference between traditional classification tasks and FSL tasks. (a) Traditional classification tasks. (b) FSL tasks.

and limit the performance and development of SAR target classification. Therefore, it is necessary to explore some models to address the problem of SAR target recognition with limited data.

One direct approach is to expand the training dataset. This can be achieved through common data augmentation techniques [21], [22], [23] and the generation of new samples using generative models [24], [25], [26]. Another effective approach is transfer learning [27], which leverages abundant data and knowledge from existing domains or tasks and transfers them to domains or tasks with limited data. By doing so, it compensates for the scarcity of data and enhances model performance. In addition, semisupervised learning [28] combines labeled and unlabeled data, making use of the information from unlabeled data to improve model performance and overcome the problem of limited data. On the other hand, unsupervised learning [29] autonomously learns the structure and patterns in data without relying on labeled information. This enables effective feature learning and model training even with limited data. However, these methods still require hundreds or even tens of training samples per class to achieve reliable SAR target classification results. Furthermore, achieving fast generalization of deep neural network models in new domains with small-scale training data remains a significant challenge, especially for SAR target images constrained by complex imaging environments.

In contrast, humans demonstrate remarkable ability in object recognition, being able to learn with very few or even no examples. Nevertheless, traditional deep learning approaches face challenges in achieving satisfactory performance when confronted with limited instances per category. This has stimulated research interest in the domains of few-shot learning (FSL) and zero-shot learning. Fe-Fei et al. [30] propose for the first time an unsupervised one-shot learning approach based on Bayesian methods. The model leverages prior knowledge and unlabeled data to overcome challenges in object category recognition, enabling effective object classification with just one example. This research offers new methods and perspectives for addressing the problem of FSL. FSL aims to recognize unknown class samples with a limited number of labeled examples, while meta-learning serves as a strategy to address the challenges of FSL. Through meta-learning [31], models can learn general learning rules and patterns, enabling them to better adapt to new tasks and learn quickly, resulting in improved performance in FSL tasks. The comparison between the training and testing sets of traditional classification tasks and FSL tasks is illustrated in Fig. 1. Generally, FSL algorithms can be classified into three categories: metric-based, model-based, and optimization-based methods. Metric-based FSL utilizes the similarity between samples to perform classification without explicitly modeling the entire data distribution. Compared to the other two types of FSL algorithms, metric-based methods often demonstrate superior generalization performance and robustness in dealing with label imbalances. In metric-based FSL, the classification of query samples is achieved by extracting their features, calculating the similarity between query and support samples, and assigning the query sample to the class of the support sample with the highest similarity.

In recent years, there emerge some metric-based FSL methods suitable for image recognition tasks, such as Siamese networks [32], matching networks [32], prototypical networks [33], graph neural networks [34], and relation networks [35]. Considering the scarcity of samples in FSL, the aforementioned algorithm based on image-level feature measurement may not be sufficiently effective. Li et al. [36] improve the matching performance between images and categories by using the image-to-class (I2C) module with local descriptors, achieving better results in FSL tasks compared to metric algorithms based on image-level features. In FSL, local descriptors often capture only the local information of an image while neglecting the global context. Due to the similarity between SAR targets and backgrounds, irrelevant local features may have an impact on the results. Therefore, relying solely on local descriptors may not fully express the image features, leading to a decrease in classification performance. This poses a challenge for SAR target recognition.

The core idea of metric-based few-shot classification methods is to measure the similarity between the feature vectors of query images and support images. Existing approaches such as ProtoNet [33] use first-order moments and Euclidean distance or cosine similarity for metric learning. Some studies have explored second-order moments [37], the combination of first and second-order moments [38] using Gaussian distributions, or introduced EMD distance [39] to calculate joint distributions. Subsequently, Xie et al. [40] proposed a deep Brownian distance covariance (DeepBDC) method for few-shot classification, which quantifies the dependencies between random variables to achieve better classification performance. In summary, the BDC metric shows great potential in few-shot image classification tasks. The algorithm calculates the prototype by averaging the BDC matrices of all samples in the support class. However, this method has some drawbacks: on one hand, it does not consider the weight differences between samples, which may overlook important information; on the other hand, for samples with noise or outliers, averaging them directly may result in an inaccurate representation of the prototype.

In the face of the aforementioned challenges, we propose a novel global and local descriptor-based metric network (GLMnet) descriptor-based for few-shot SAR target classification. Considering the influence of unrelated local features due to the similarity between SAR targets and backgrounds, using only local descriptors may not fully capture the image features, thus affecting the classification performance. To address this issue, we introduce the BDC metric module to capture the global relationships between local descriptors. Furthermore, to better integrate the BDC metric module and the I2C module, we design an adaptive metric fusion module (AMFM) to achieve discriminative metric performance. Additionally, when computing the prototypes of support classes, the BDC metric module adopts an averaging approach over the BDC matrices

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:45:04 UTC from IEEE Xplore. Restrictions apply.

of all samples in that class. However, this method does not consider the weight differences between samples and may lead to inaccurate prototype representation. Therefore, we design a weighted average prototype computation module (WAPCM) to obtain accurate prototypes for each class and improve the classification performance of SAR targets.

The key contributions of this article can be summarized as follows.

1) Despite the fact that using local descriptors can provide more information compared to image-level representations, they typically capture only local information of the image, missing out on global contextual information and potentially failing to fully express the features of the images. Additionally, due to the similarity between SAR targets and the background, irrelevant local features may impact the classification performance of SAR targets. To address this issue, we introduce the BDC metric module, which considers the global relationships among local descriptors. In this way, by combining the global relationships (BDC metric module) and local relationships (I2C metric module), we measure the similarity between the feature vectors of query samples and support classes.
2) To better integrate the BDC metric module and the I2C metric module, we designed an AMFM. This module learns a trainable 2-D weight vector to adaptively fuse the two metric relationships, aiming to achieve discriminative metric performance.
3) Due to the BDC metric module using the average of the BDC matrices of all samples in the support class to compute the prototype, this method fails to consider the weight differences among samples, which may result in inaccurate prototype representation and impact the classification performance of the targets. Therefore, we designed a WAPCM that takes into account the weight differences among samples in the support class, aiming to obtain accurate prototypes for each class and improve the classification performance of SAR targets.

The remaining sections of this article are organized as follows. Section II reviews FSL SAR target classification algorithms and metric-based FSL algorithms. Section III provides a brief description of the task settings and training strategies related to FSL. Section IV introduces the specific framework of the proposed GLMnet. Section V presents the experimental results and analysis. Finally, Section VI presents the conclusion.

II. Related Work

A. SAR Target Classification Algorithms Based on FSL

SAR target classification is an important research field that aims to automatically identify and classify targets in SAR images. In CNN-based SAR target classification algorithms, a large amount of annotated data are typically required for training to establish accurate classification models. However, acquiring a massive quantity of annotated data is an expensive and time-consuming task, especially in the domain of SAR images. Few-shot classification aims to adjust classifiers using limited annotated data to adapt to novel classes that were not seen during training. Recently, researchers have started focusing on SAR target classification algorithms based on FSL. Wang et al. [41] propose a hybrid inference network for few-shot SAR ATR, providing new research perspectives and methods for FSL in the field of SAR target recognition. To address the issue of limited SAR samples, a meta-learning framework, a GNN network based on mixed loss [42], and a transductive prototype attention reasoning network [43] are proposed. Additionally, Zhang et al. [44] present an innovative network called DKTS-N, which integrates knowledge from azimuth, amplitude, and phase data in the SAR domain, pioneering the few-shot SAR vehicle recognition. An improved prototype network [45] based on spatial transformation is also proposed. Furthermore, Zhang et al. [20] propose a novel meta-transfer learning method for cross-task and cross-domain SAR target recognition. Moreover, to achieve continuous learning of new classes with limited training samples, Wang et al. [46] present a few-shot incremental SAR ATR method.

B. Metric-Based FSL Algorithms

Metric-based FSL approaches achieve effective classification or recognition by measuring the similarity between query samples and support classes. Siamese neural networks [32] are introduced as a method to address the one-shot problem, extending the network's predictive capabilities to new data using their powerful discriminative ability. Matching networks [47] map a small labeled support set and an unlabeled query example to their network labels, eliminating the need for fine-tuning on new unknown types. Importantly, this approach proposes an episode-based training strategy that is crucial for further advancing FSL. Prototype networks [33] offer a simple and effective method by computing prototypes of support classes (the average of feature vectors of support class samples) and determining the class membership of query samples based on the Euclidean distance between them and each support class prototype. Relation networks [35] are another method used for learning relationships between samples and classifying them, taking the feature representations of samples as input and learning a function that combines the features of two samples to determine if they belong to the same category. Due to the limited number of samples in FSL, previous metric algorithms based on image-level features may not be sufficiently effective. To improve metric performance, Li et al. [36] proposed a method called deep nearest neighbor network (DN4). DN4 utilizes an I2C metric model with local descriptors, significantly improving the metric performance of the network. DN4 demonstrates satisfactory image classification results in FSL tasks.

The main idea of metric-based few-shot classification is to learn feature representations through deep convolutional networks and determine the class membership by measuring the similarity between query images and support class images. From a statistical perspective, the features of query and support class images can be seen as observations of random vectors in a high-dimensional embedding space. Based on this view, probability distributions can be used to measure the similarity between query images and support class images. However, modeling high-dimensional feature distributions is challenging, and a common approach is to model statistical moments. ProtoNet [33] represents images using the first moment (the average of feature vectors of support class images) and uses Euclidean distance for similarity computation. To obtain

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:45:04 UTC from IEEE Xplore. Restrictions apply.

richer statistical information, some methods have explored second moments (covariance matrices) [37] or combinations of first and second moments [38], using Frobenius norm or Kullback–Leibler (KL) divergence as similarity measures. However, these methods only utilize marginal distributions and ignore joint distributions, limiting the model's performance. Additionally, covariance can only model linear relationships, while the dependencies between query and support class images should be measured based on their joint distribution. EMD [39] is an effective method for measuring such correlations by seeking optimal joint distributions. However, this method is computationally expensive and lacks real-time performance. To address this issue and fully exploit the distributional relationships within samples, Xie et al. [40] proposed the BDC metric algorithm, which measures the covariance of Brownian motion distances between samples and classes, resulting in more accurate distance metrics. This model can handle nonlinear relationships freely and has analytical and computational efficiency. This also demonstrates the tremendous potential of the BDC metric in FSL.

The DN4 algorithm achieves good image classification performance in FSL by utilizing local descriptors. However, local descriptors only capture the local information of an image and overlook global relationships. Additionally, when calculating similarities, irrelevant local features may influence the classification results due to the similarity between SAR targets and backgrounds. Therefore, relying solely on local descriptors may not fully express the image features, leading to a decline in classification performance. On the other hand, although the BDC metric demonstrates significant advantages in this algorithm, there are two drawbacks in prototype computation. First, it does not consider the weight differences between samples, potentially overlooking the influence of important samples on the prototypes. Second, for samples that contain noise or outliers, simply averaging them may result in an inaccurate representation of prototypes, thereby affecting the metric capabilities of the model.

III. Preliminary

A. Problem Formulation

The few-shot classification task consists of three datasets: the auxiliary set $\mathcal{A}$, the support set S, and the query set Q. The support set S and the query set Q share the same label space, similar to the training and testing sets in general image classification tasks. The goal of FSL is to classify the unlabeled samples in the query set Q based on the support set S. Assuming that the support set S contains C categories and each category has K samples, the few-shot classification task is called N-way K-shot [47]. However, due to the extremely limited number of samples per category in the support set S (e.g., only 1 or 5), deep neural networks often suffer from overfitting and are challenging to train effectively. To address this issue, the auxiliary set $\mathcal{A}$ is usually utilized to learn transferable meta-knowledge that enables the model to adapt to new tasks with unseen classes that have only a few samples. Moreover, the auxiliary set $\mathcal{A}$ contains numerous categories and labeled samples, and its label space does not intersect with the support set S. Therefore, the labeled information can be fully exploited to enhance the generalization ability of the model.

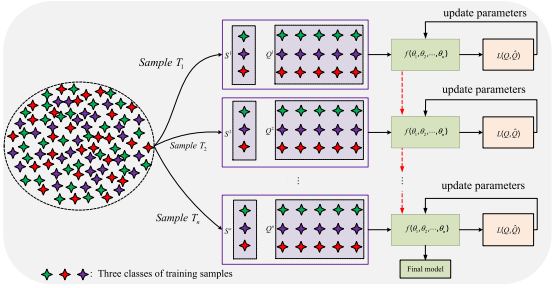


Fig. 2. Demonstration of three-way one-shot SAR target classification setup.

B. Episodic Training

In SAR target recognition, it is difficult to train a model with strong generalization ability due to the limited sample data. To address this problem, we adopt an episode training mechanism, which has been proven to be notably effective in other metric-based FSL algorithms. Specifically, we randomly construct each episode (task) T from the auxiliary set $\mathcal{A}$ according to the N-way K-shot strategy to simulate the few-shot task. In each iteration, the first task uses the loss function $L(Q, \hat{Q})$ to update the parameters and train the current model $f(\theta_1, \theta_2, \ldots, \theta_n)$. Then, the trained model is used as the initialization for the next task to continue model training. During the training process, a large number of tasks will be randomly constructed, and after training is completed, the trained model based on S will be used to predict the labels of Q. The training process is shown in Fig. 2.

IV. Proposed Method

The proposed GLMnet consists of three components: an embedding module, a discriminative metric module, and a classifier module. First, we input the support images and query images into the embedding module to generate their feature vectors. The discriminative metric module comprises global-level and local-level metric relationships. Specifically, the global-level metric relationships capture the relationships between image blocks using the BDC metric module. The local-level metric relationships are calculated by the I2C module. Next, we design an AMFM as the final classifier to classify the query images. Furthermore, we design a WAPCM to compute accurate prototypes for each class. Finally, we train the model end-to-end using the cross-entropy loss function. The architecture of GLMnet is illustrated in Fig. 3, and the specific framework details will be described in the following text.

A. Embedding Module

Mapping the support sample x_i and query sample q_i into the embedding module $f_\vartheta(\cdot)$, we can obtain the output feature vectors $f_\vartheta(x_i)$ and $f_\vartheta(q_i)$. In short, given an image x, $f_\vartheta(x)$ is a $c \times h \times w$ tensor, which can be regarded as a set of c-dimensional local descriptors

$$f_\vartheta(x) = [x_1, x_2, \ldots, x_m] \in \mathbb{R}^{c \times m} \tag{1}$$

where x_i represents the ith local descriptor, and $m = h \times w$ is the total number of local descriptors in image x. These local descriptors indicate the local representations of spatial patches in the image.

The architecture of the embedding network, as shown in Fig. 4, consists of four layers. The initial two layers of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:45:04 UTC from IEEE Xplore. Restrictions apply.

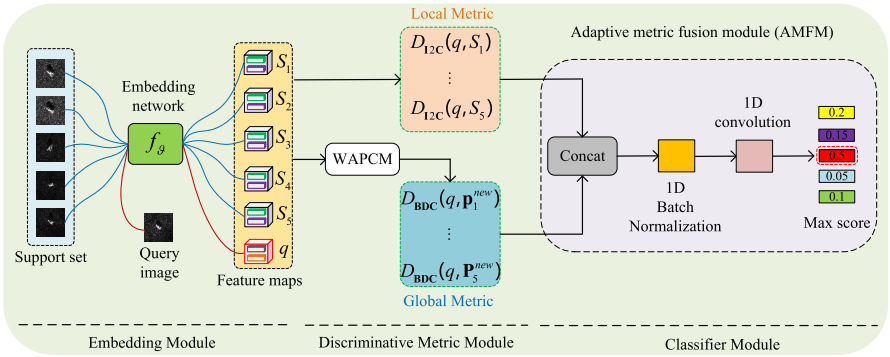


Fig. 3. Architecture of the proposed GLMnet in the five-way one-shot setting.

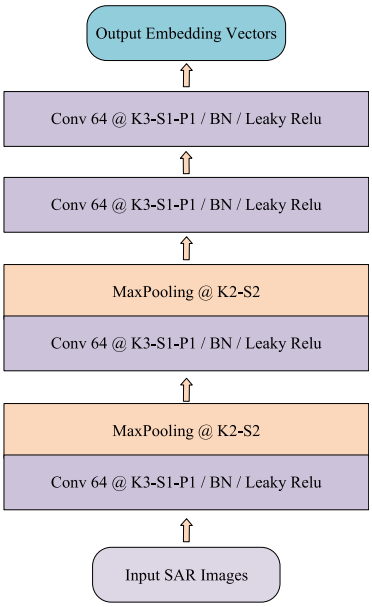


Fig. 4. Structure of the embedding network.

the embedding network consist of convolution layers and max-pooling layers, while the last two layers only include convolution layers. Specifically, the notation "Conv64@K3-S1-P1/BN/Leaky Relu" indicates a convolution operation with 64 kernels of size 3×3, a stride of 1, and a padding value of 1. Additionally, "BN" refers to batch normalization, a technique that normalizes a set of data with a large range gap or different units to a specified range based on specific rules [48]. Furthermore, the Leaky ReLU activation function [49] is utilized to enhance the model's robustness, which can lead to better classification performance compared to the ReLU activation function [50]. On the other hand, the notation "MaxPooling@K2-S2" signifies max-pooling with a kernel size of 2×2 and a stride of 2.

B. Discriminative Metric Module

1) I2C Metric Module: DN4 is a classifier based on the naive Bayes nearest neighbor approach, which employs the I2C metric for classification. This method directly calculates the I2C distance using the nearest neighbor distances in the space of local image descriptors, without the need for quantization of the descriptors. Traditional quantization methods may lead to the loss of discriminative information due to quantization errors, and this loss is irreparable, resulting in decreased classification performance, especially in FSL scenarios. In contrast, DN4 utilizes the I2C metric, directly computing the distances between the descriptors of query images and the entire class, effectively alleviating the issue of quantization errors. Previous research has mainly focused on feature representations at the image level, but due to insufficient samples, image-level feature representations cannot effectively estimate the distribution of classes. The I2C metric based on local descriptors exhibits stronger generalization ability compared to the measure based on image-level features.

Specifically, through the embedding module $f_\vartheta(\cdot)$, a query image q and a support class S can be embedded as $f_\vartheta(q) = [x_1, x_2, \ldots, x_m] \in \mathbb{R}^{c\times m}$ and $f_\vartheta(S) = [f_\vartheta(X_1), f_\vartheta(X_2), \ldots, f_\vartheta(X_K)] \in \mathbb{R}^{c\times mK}$, where K represents the number of support samples in S. From a mathematical perspective, the I2C similarity measure, concerning the query image q and the support class S, can be represented as

$$D_{\mathrm{I2C}}(q, S) = \sum_{i=1}^{m}\sum_{j=1}^{k} \cos\left(x_i, \widehat{x}_i^{\,j}\right) \tag{2}$$

where $\widehat{x}_i^{\,j}|_{j=1}^{k}$ represents the k nearest neighbors of the descriptor x_i found within the class S. Typically, k is set to 3 in this article. $\cos(\cdot\cdot)$ represents the cosine similarity function, which can be expressed as

$$\cos\left(x_i, \widehat{x}_i\right) = \frac{x_i^{\top}\widehat{x}_i}{\left\|x_i\right\| \cdot \left\|\widehat{x}_i\right\|}. \tag{3}$$

For each support class, we gather the local descriptors of all the images in that class to estimate their distribution in the feature space. Local descriptors capture subtle local information in the images, making them crucial for the final image recognition task. By combining the local descriptors of all the images, we gain a more comprehensive understanding of the distribution of the support classes in the feature space. This approach helps us better capture the differences and similarities between classes, enhances the modeling capability of the support classes, and thus improves the performance of few-shot image recognition.

2) BDC Metric Module: The I2C metric achieves good image classification performance in FSL by utilizing local descriptors. However, local descriptors only capture the local information of an image and neglect the global information between images. Therefore, relying solely on local descriptors

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:45:04 UTC from IEEE Xplore. Restrictions apply.

may not fully represent the image features. Additionally, due to the similar spatial backgrounds of SAR targets, irrelevant local features (i.e., the background) may impact the classification results of the targets when calculating similarity. Moreover, due to the limited number of samples, each nearest neighbor descriptor contains limited information, leading to a lack of robustness in handling background noise in the model. To address this problem, we introduce a method called BDC. BDC is an approach for learning image representations by comparing the differences between the joint feature function of embedded features and the product of marginal values, enabling the capture of global relationships among local descriptors.

The BDC theory, established in reference from the perspective of feature functions, computes the Euclidean distance between the joint feature function of two random vectors and the product of their marginal distributions. For a given p-dimensional random vector $X \in \mathbb{R}^p$ and q-dimensional random vector $Y \in \mathbb{R}^q$, the BDC metric can be expressed as

$$\xi(X,Y) = \int_{\mathbb{R}^p}\int_{\mathbb{R}^q} \frac{|\phi_{XY}(\mathbf{u},\mathbf{v}) - \phi_X(\mathbf{u})\phi_Y(\mathbf{v})|^2}{c_p c_q \|\mathbf{u}\|^{1+p}\|\mathbf{v}\|^{1+q}} d\mathbf{u}d\mathbf{v} \quad (4)$$

where $\|\cdot\|$ represents the Euclidean norm, $c_p = \pi^{(1+p)/2}/\Gamma((1+p)/2)$, where Γ is the complete gamma function. The marginal distributions of X and Y are denoted as $\phi_X(\mathbf{u}) = \phi_{XY}(\mathbf{u},\mathbf{0})$ and $\phi_Y(\mathbf{v}) = \phi_{XY}(\mathbf{0},\mathbf{v})$, respectively, where $\mathbf{0}$ represents a vector of all zeros. The joint feature function $\phi_{XY}(\mathbf{u},\mathbf{v})$ of X and Y can be expressed as

$$\phi_{XY}(\mathbf{u},\mathbf{v}) = \int_{\mathbb{R}^p}\int_{\mathbb{R}^q} \exp\left(i\left(\mathbf{u}^T\mathbf{x} + \mathbf{v}^T\mathbf{y}\right)\right) f_{XY}(\mathbf{x},\mathbf{y}) d\mathbf{x}d\mathbf{y} \quad (5)$$

where i represents the imaginary unit, and $f_{XY}(\mathbf{x},\mathbf{y})$ represents the probability density function of X and Y. According to (4) and (5), the BDC metric takes into account both the joint distribution and the marginal distribution of two random variables.

Generally, it is not possible to know the probability distribution functions of each random variable. However, we can obtain a set of observations for each random variable. Assuming that the variables X and Y have a set of observations, denoted as $\{\mathbf{x}_1,\ldots,\mathbf{x}_b\}$ and $\{\mathbf{y}_1,\ldots,\mathbf{y}_b\}$, respectively. Furthermore, all the observed pairs $\{(\mathbf{x}_1,\mathbf{y}_1),\ldots,(\mathbf{x}_b,\mathbf{y}_b)\}$ are independently and identically distributed. In this case, we can define $\phi_{XY}(\mathbf{u},\mathbf{v})$ as

$$\phi_{XY}(\mathbf{u},\mathbf{v}) = \frac{1}{b}\sum_{i=1}^{b} \exp\left(i\left(\mathbf{u}^T\mathbf{x}_i + \mathbf{v}^T\mathbf{y}_i\right)\right). \quad (6)$$

For discrete observations, the BDC metric can be expressed in a discrete form. Let $\hat{R} = (\hat{r}_{mn}) \in \mathbb{R}^{b\times b}$, where $\hat{r}_{mn} = \|\mathbf{x}_m - \mathbf{x}_n\|$ represents the Euclidean distance matrix computed from the discrete observations of X. Similarly, we can obtain the Euclidean distance matrix $\hat{S} = (\hat{s}_{mn}) \in \mathbb{R}^{b\times b}$ for Y, where $\hat{s}_{mn} = \|\mathbf{x}_m - \mathbf{x}_n\|$. In this case, the BDC matrix can be expressed as

$$\chi(X,Y) = \frac{1}{b^2}\mathrm{tr}\left(\mathbf{R}^T\mathbf{S}\right) \quad (7)$$

where tr $(\cdot)$ represents the trace of a matrix and T denotes matrix transpose. $\mathbf{R} = (r_{mn})$ represents the BDC matrix of X. Here, $r_{mn} = \hat{r}_{mn} - (1/b)\sum_{m=1}^{b}\hat{r}_{mn} - (1/b)\sum_{n=1}^{b}\hat{r}_{mn} - (1/b^2)\sum_{m=1}^{b}\sum_{n=1}^{b}\hat{r}_{mn}$, where the last three terms correspond to the mth column, nth row, and the average of all entries of $\hat{R}$, respectively. Similarly, matrix $\mathbf{S}$ can be computed from $\hat{S}$. Since $1/b^2$ is a constant and the BDC matrix is symmetric, $\chi(X,Y)$ can also be expressed as the inner product of two BDC vectors $\mathbf{r}$ and $\mathbf{s}$

$$\chi(X,Y) = <\mathbf{r},\mathbf{s}> = \mathbf{r}^T\mathbf{s} \quad (8)$$

where $\langle\cdot\rangle$ represents matrix inner product. $\mathbf{r}$ and $\mathbf{s}$ are obtained by extracting the upper triangular parts of $\mathbf{R}$ and $\mathbf{S}$, and then performing vectorization. It is worth noting that the similarity between $\mathbf{R}$ and $\mathbf{S}$, can also be measured using Euclidean distance.

According to (7) and (8), it can be observed that the BDC metric can be computed independently for each input image, yielding separate BDC matrices. Specifically, given the embedding feature $\mathbf{A} \in \mathbb{R}^{h\times w\times d}$, which is reshaped to $\mathbf{D} \in \mathbb{R}^{hw\times d}$, each column $r \in R^{hw}$ can be regarded as a discrete observation of $\mathbf{D}$. Mathematically, the BDC matrix $\mathbf{R}$ can be computed using the following equation:

$$\tilde{R} = 2\left(\mathbf{1}\left(\mathbf{D}^T\mathbf{D}\circ\mathbf{I}\right)\right)_{\mathrm{sym}} - 2\mathbf{D}^T\mathbf{D} \quad (9)$$

$$\hat{R} = \left(\sqrt{\tilde{r}_{mn}}\right) \quad (10)$$

$$\mathbf{R} = \hat{R} - \frac{2}{b}\left(\mathbf{1}\hat{R}\right)_{\mathrm{sym}} + \frac{1}{b^2}\mathbf{1}\hat{R}\mathbf{1} \quad (11)$$

where $\tilde{R} = (\tilde{r}_{mn})$ and $\tilde{r}_{mn}$ represents the Euclidean distance between the mth column and the nth column. Additionally, $\mathbf{1} \in \mathbb{R}^{b\times b}$ denotes a matrix with all elements equal to 1. $\circ$ denotes the Hadamard product operation between matrices, $\mathbf{I}$ represents the identity matrix, and $(\mathbf{Z})_{\mathrm{sym}} = (1/2)(\mathbf{Z}+\mathbf{Z}^T)$. According to (9)–(11), we can observe that in computing the BDC matrix of sample features, each row of feature descriptors for the sample features and each column of feature descriptors are both involved in the computation. Therefore, we believe that the global-level metric relationship based on BDC considers the global relationship between local descriptors.

The prototype of the support class S can be represented as the average of the BDC matrices of all the sample features belonging to that class

$$\mathbf{P}_k = \frac{1}{K}\sum_{i=1}^{K}\mathbf{R}_\theta(S_i) \quad (12)$$

where $\mathbf{R}_\theta(\cdot)$ represents the BDC module and K represents the number of samples in the support class S.

The metric based on BDC with respect to the support class S and the query sample q can be expressed as

$$D_{\mathrm{BDC}}(q,S) = d(\mathbf{R}_\theta(q),\mathbf{P}_k) \quad (13)$$

where d represents the Euclidean distance.

C. Classifier Module

The I2C metric based on local descriptors only captures the local information of the image, ignoring global information, thus it cannot fully represent the image features. Additionally, due to the spatial background similarity of SAR targets, some irrelevant local features are detrimental to the classification

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:45:04 UTC from IEEE Xplore. Restrictions apply.

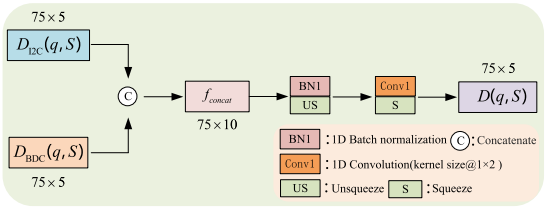


Fig. 5. Structure of the AMFM.

results. In contrast, the BDC metric takes into account the differences between the joint feature function and the marginal values to learn image representations, allowing for a comprehensive capture of the global relationships between local descriptors. Therefore, a fusion strategy is needed to integrate these two metric relationships, namely the local metric relationship based on I2C and the global metric relationship based on BDC. To address this issue, we design an AMFM that includes a learnable 2-D weight vector $w = [w1, w2]$ to achieve more discriminative metric performance. Specifically, the final fused similarity between the query sample q and the support class S can be defined as

$$D(q, S) = D_{\text{I2C}}(q, S) \cdot w_1 + D_{BDC}(q, \mathbf{P}_k) \cdot \omega_2. \tag{14}$$

For a five-way one-shot task and a specific query q, as illustrated in Fig. 5, the I2C branch and BDC branch output a 5-D similarity vector each. These two similarity vectors are concatenated to form a 10-D vector, which is then passed through a 1-D convolutional layer with a kernel size of 1×1 and a dilation value of 5. This produces a 5-D weighted similarity vector, learned using a 2-D weight ω. To balance the similarity scale between the two parts, a BN layer is added before the 1-D convolutional layer. The final classification result is obtained through a nonparametric nearest neighbor classifier.

D. Weighted Average Prototype Computation Module

The BDC metric calculates the prototype by averaging the BDC matrices of all samples in that category. However, this approach does not consider the weight differences between samples, which may overlook some important information. Additionally, for samples that contain noise or outliers, directly averaging them may result in an inaccurate representation of the prototype, further affecting the accuracy of the metric.

Vaswani et al. [51] propose the Transformer, employing a self-attention mechanism. This allows the model to adaptively weigh each position in the input sequence, capturing long-distance dependencies and enhancing sequence modeling capabilities. Inspired by the weight calculation in the Transformer, we propose a WAPCM, as demonstrated in Fig. 6 under two-shot setting. This module considers the BDC matrix weights for the support class samples to obtain a more accurate support class prototype. In the Transformer, each word is assigned three different vectors: the query vector $\mathbf{Q}$, the key vector $\mathbf{K}$, and the value vector $\mathbf{V}$. These vectors are obtained by multiplying the embedding vector X with three different weight matrices W_Q, W_K, and W_V, where the dimensions of these three matrices are the same. However, in FSL, the number of support samples per task is very limited, and introducing three weight matrices may lead to overfitting. Furthermore, the BDC matrix of the support class samples

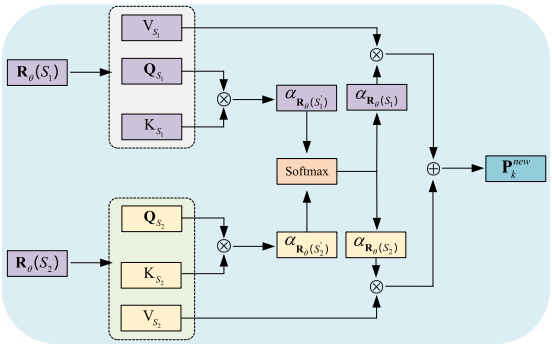


Fig. 6. Structure of the WAPCM.

itself contains certain semantic information. Taking all these factors into consideration, we use the BDC matrix $\mathbf{R}_\theta(S_i)$ of the support class samples as their query vector $\mathbf{Q}$, key vector $\mathbf{K}$, and value vector $\mathbf{V}$, which can be represented as

$$\mathbf{Q}_{S_i} = \mathbf{K}_{S_i} = \mathbf{V}_{S_i} = \mathbf{R}_\theta(S_i) \tag{15}$$

where $\mathbf{R}_\theta(S_i)$ represents the BDC matrix of the ith sample in the support class S.

In this article, we choose to use the average of the BDC matrices of all support samples as the query vector. This is because the average operation can summarize the information of the feature vectors of support samples and preserve their overall characteristics. According to (12), the query vector of the BDC matrix for the samples in the support class S can be represented as

$$\mathbf{Q}_{S_i} = \mathbf{P}_k. \tag{16}$$

The weight $\alpha_{\mathbf{R}_\theta(S_i)}$ of the BDC matrix $\mathbf{R}_\theta(S_i)$ for the support class sample S_i can be represented as

$$\alpha_{\mathbf{R}_\theta(S_i)} = \frac{\exp\left(\mathbf{Q}_{S_i}{}^T \mathbf{K}_{S_i}\right)}{\sum_{j=1}^{K} \exp\left(\mathbf{Q}_{S_j}{}^T \mathbf{K}_{S_j}\right)}. \tag{17}$$

The prototype of the support class S after weighted averaging, denoted as $\mathbf{P}_k^{\text{new}}$, can be represented as

$$\mathbf{P}_k^{\text{new}} = \sum_{i=1}^{K} \alpha_{\mathbf{R}_\theta(S_i)} \mathbf{V}_{S_i}. \tag{18}$$

By doing so, we have obtained a new prototype $\mathbf{P}_k^{\text{new}}$ for the support class S, which takes into account the weight differences between the BDC matrices of the support samples. This further improves the performance of the BDC metric.

V. Experiments Results

A. Datasets Description

The MSTAR dataset is jointly funded and established by the Defense Advanced Research Projects Agency (DARPA) and the Air Force Research Laboratory (AFRL). The dataset is the most commonly used in the field of SAR ATR. It contains SAR image slices collected from X-band airborne SAR imaging. The experimental data has a depression angle of 17°. The radar has a resolution of 0.3 × 0.3 m, and the images have a size of 128 × 128. The MSTAR dataset consists of ten military target classes, including T-62, 2S1, BTR-70, T-72, ZIL-131,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:45:04 UTC from IEEE Xplore. Restrictions apply.

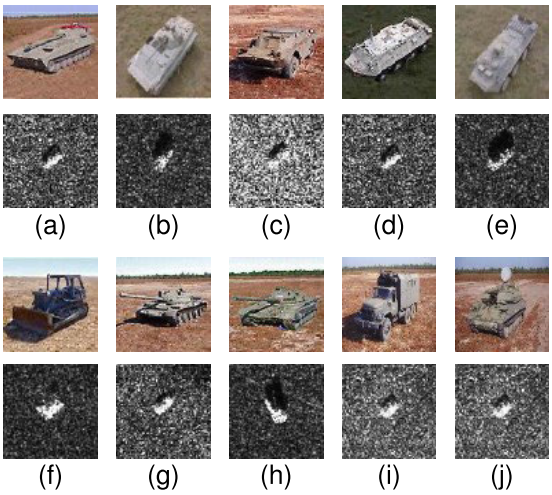


Fig. 7. (Top) Optical images and (bottom) SAR images with 17° depression angle in MSTAR dataset. (a) 2S1. (b) BMP-2. (c) BRDM-2. (d) BTR-60. (e) BTR-70. (f) D-7. (g) T-62. (h) T-72. (i) ZIL-131. (j) ZSU-234.

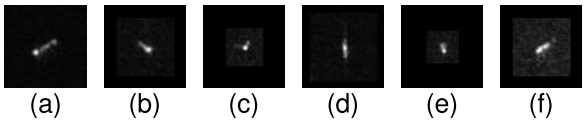


Fig. 8. Six classes of ship targets in the OpenSARShip dataset. (a) Cargo. (b) Tanker. (c) Tug. (d) Dredging. (e) Fishing. (f) Passengers.

BRDM-2, D-7, BMP-2, BTR-60, and ZSU-234. Fig. 7 illustrates the SAR images of military targets from the MSTAR dataset along with their corresponding optical images.

The OpenSARShip dataset [52] is collected by Shanghai Jiao Tong University in 2017. This dataset includes sliced data of 11 346 ships, covering 17 types of Automatic Identification Systems (AISs). The sliced data were acquired using the VH and VV polarization modes of the Sentinel-1A satellite. Each sliced image has a size of 64 × 64. Fig. 8 illustrates six classes of ship target data from the OpenSARShip dataset.

In recent years, several datasets related to SAR have emerged, including AIR-SARShip [53] and SAR-AIRcraft [54]. The AIR-SARShip dataset comprises only one category, making it unsuitable for target classification experiments. The SAR-AIRcraft dataset is obtained from the GF-3 satellite and employs a coherent imaging mode with single polarization. The images have a resolution of 1 m and consist of four different sizes: 800 × 800, 1000 × 1000, 1200 × 1200, and 1500 × 1500. Furthermore, the dataset contains 4368 images, covering 16 463 instances of aircraft targets. It includes seven categories: A220, A320/321, A330, ARJ21, Boeing737, Boeing787, and others. Fig. 9 illustrates the SAR images of aircraft targets from the SAR-AIRcraft dataset along with their corresponding optical images. It can be used for target detection tasks, target recognition tasks, and combined target detection and recognition tasks. The characteristics of the dataset encompass complex scenes, diverse target categories, dense target distribution, noise interference, diverse task types, and multiscale attributes.

The partitioning of training data D_{train} and testing data D_{test} for three SAR datasets is described as follows.

1) MSTAR Dataset: The training dataset D_{train} of the MSTAR dataset includes the classes 2S1, BMP-2, BRDM-2, BTR-60, and BTR-70. The test dataset D_{test} includes the classes D-7, T-62, T-72, ZIL-131, and ZSU-234. For detailed classification information, please refer to Table I.

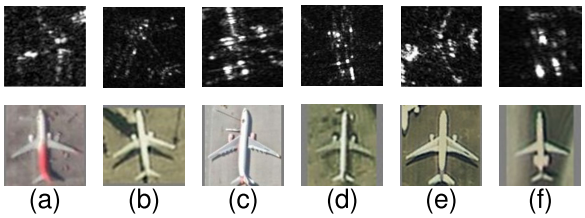


Fig. 9. (Top) SAR images and (bottom) optical images in the SAR-AIRcraft dataset. (a) A220. (b) Boeing737. (c) Boeing787. (d) A320/321. (e) A330. (f) ARJ21.

TABLE I
DIVISION OF TRAINING DATA AND TESTING DATA IN MSTAR DATASET

	Class	Depression	Num
D_{train}	BMP-2	17°	232
	BTR-70	17°	233
	2S1	17°	299
	BRDM-2	17°	298
	BTR-60	17°	256
D_{test}	T62	17°	299
	D-7	17°	299
	ZSU-234	17°	299
	T-72	17°	232
	ZIL-131	17°	299

TABLE II
DIVISION OF TRAINING DATA AND TESTING DATA IN OPENSARSHIP DATASET

	Class	Num
D_{train}	Cargo	222
	Tanker	240
	Tug	260
D_{test}	Dredging	80
	Fishing	80
	Passengers	80

2) OpenSARShip Dataset: Considering that the number of images in most categories is relatively small, we select six categories for the training dataset D_{train} (i.e., Cargo, Tanker, and Tug) and the test dataset D_{test} (i.e., Dredging, Fishing, and Passengers), as shown in Table II. In the training dataset, the Tug category had only 80 available samples. To ensure the balance of the training data, we applied data augmentation techniques to the Tug category, including adding noise, increasing brightness, and performing rotations. This resulted in a total of 260 or the Tug category.

3) SAR-AIRcraft Dataset: We select six categories for the training dataset D_{train} (i.e., A220, Boeing737, and Boeing787) and the test dataset D_{test} (i.e., A320/321, A330, and ARJ21), as shown in Table III.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:45:04 UTC from IEEE Xplore. Restrictions apply.

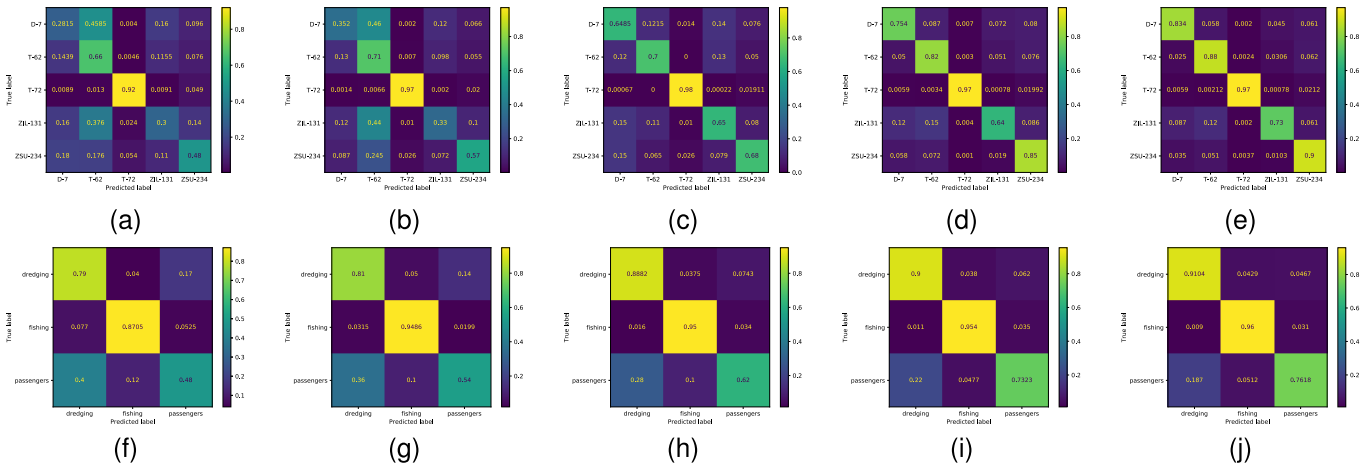


Fig. 10. Confusion matrix of the proposed GLMnet under different settings on (a)–(e) MSTAR and (f)–(j) OpenSARship datasets. (a) five-way one-shot. (b) five-way two-shot. (c) five-way five-shot. (d) five-way ten-shot. (e) five-way 15-shot. (f) three-way one-shot. (g) three-way two-shot. (h) three-way five-shot. (i) three-way ten-shot. (j) three-way 15-shot.

TABLE III
DIVISION OF TRAINING DATA AND TESTING DATA IN SAR-AIRCRAFT DATASET

	Class	Num
D_{train}	A220	1000
	Boeing737	1000
	Boeing787	1000
D_{test}	A320/321	200
	A330	200
	ARJ21	200

The experiment is conducted using the NVIDIA GeForce GTX 1650 GPU with PyTorch version 1.8 and Python version 3.7. The learning rate and momentum parameters are set to 0.001 and 0.5, respectively. We employ the Adam optimizer and cross-entropy loss function for training. The training phase consists of 30 epochs, with each epoch containing 10 000 episodes. Additionally, 600 episodes are conducted during the testing phase.

Finally, it is worth noting that the classes in the training dataset D_{train} and the testing dataset D_{test} do not overlap on these three SAR datasets. In other words, there is no intersection between D_{train} and D_{test}.

B. Target Classification Performance on Three SAR Datasets

It is worth noting that each task is performed by randomly sampling support samples and query samples, so the classification accuracy of each task may vary during the testing process. To obtain relatively accurate classification accuracy, we conducted tests on 600 tasks, where each task randomly selected 15 samples per category as query samples. Table IV shows the classification accuracy of the proposed GLMnet on three datasets, MSTAR (five-way), OpenSARship (three-way), and SAR-AIRcraft (three-way) under different shot settings. The proposed GLMnet achieves the highest classification accuracies of 86.28%, 87.74%, and 78.66% on datasets MSTAR (five-way 15-shot), OpenSARShip (three-way 15-shot), and SAR-AIRcraft (three-way 15-shot), respectively. It is evident that as the shot increases, the classification accuracy also increases. This is because each category has more samples available for learning, enhancing the richness of the data and allowing the model to better learn the features of each category, thereby improving accuracy.

To provide a clearer visualization of the classification performance of the proposed GLMnet on each category, we used confusion matrices to visualize its performance on the MSTAR and OpenSARship datasets, shown in Fig. 10. The experimental results show that our proposed algorithm performs well in terms of classification performance.

C. Ablation Study

1) Effectiveness of AMFM: To validate the effectiveness of AMFM, we conduct experiments on datasets MSTAR, OpenSARship, and SAR-AIRcraft, as shown in Tables V–VII. We use a five-way setting for the MSTAR dataset and a three-way setting for the other two datasets (OpenSARship and SAR-AIRcraft). From the experimental results, it is evident that the classification accuracy using AMFM is superior to that using the I2C metric module. Specifically, in the five-way five-shot setting of the MSTAR dataset and the three-way one-shot setting of the OpenSARship dataset, the classification accuracy using AMFM is improved by 1.9% and 1.07% compared to the I2C metric module, respectively. It is worth noting that the accuracy increases by 6.37% on the SAR-AIRcraft dataset under the setting of three-way one-shot. The above results indicate that AMFM can effectively constrain the embedding network to learn embedding vectors with better separability.

To provide a more intuitive explanation for the effectiveness of AMFM, we further validate the effectiveness of the discriminative metric module using t-SNE. As shown in Fig. 11, the embedding vectors in the embedding space generated by the discriminative metric module exhibit better class separability and intraclass compactness compared to those based on the I2C metric. Compared to the I2C metric module, our proposed AMFM can more effectively concentrate the distribution of embedding vectors belonging to the same target, making it easier to define reasonable decision boundaries. In conclusion,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:45:04 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

CLASSIFICATION ACCURACY OF THE GLMNET ALGORITHM ON DATASETS MSTAR (5)-WAY), OPENSARSHIP (THREE-WAY) AND SAR-AIRCRAFT UNDER DIFFERENT SETTINGS

Dataset	Accuracy(%)				
	1-shot	2-shot	5-shot	10-shot	15-shot
MSTAR	52.83±0.33	58.64±0.27	73.17±0.16	80.68±0.24	86.28±0.12
OpenSARship	71.35±0.32	76.62±0.33	81.94±0.23	86.21±0.17	87.74±0.18
SAR-AIRcraft	46.42±0.46	54.72±0.41	61.95±0.38	71.49±0.21	78.66±0.16

TABLE V

CLASSIFICATION ACCURACY COMPARISON USING DIFFERENT METHODS ON THE MSTAR DATASET (FIVE-WAY) UNDER DIFFERENT SETTINGS

Method	Accuracy(%)				
	1-shot	2-shot	5-shot	10-shot	15-shot
I2C	52.32±0.46	57.03±0.37	68.68±0.23	79.28±0.12	84.79±0.21
AMFM	**52.83±0.33**	**57.68±0.14**	**70.58±0.42**	**80.02±0.28**	**85.74±0.18**
AMFM+WAPCM	-	**58.64±0.27**	**73.17±0.16**	**80.68±0.24**	**86.28±0.12**

TABLE VI

CLASSIFICATION ACCURACY COMPARISON USING DIFFERENT METHODS ON THE OPENSARSHIP DATASET (THREE-WAY) UNDER DIFFERENT SETTINGS

Method	Accuracy(%)				
	1-shot	2-shot	5-shot	10-shot	15-shot
I2C	70.28±0.25	72.19±0.21	79.81±0.23	85.26±0.37	86.19±0.22
AMFM	**71.35±0.32**	**73.19±0.16**	**80.85±0.12**	**85.73±0.22**	**86.92±0.15**
AMFM+WAPCM	-	**76.62±0.33**	**81.94±0.23**	**86.21±0.17**	**87.74±0.18**

TABLE VII

CLASSIFICATION ACCURACY COMPARISON USING DIFFERENT METHODS ON THE SAR-AIRCRAFT DATASET (THREE-WAY) UNDER DIFFERENT SETTINGS

Method	Accuracy(%)				
	1-shot	2-shot	5-shot	10-shot	15-shot
I2C	40.05±0.60	49.97±0.22	60.74±0.28	70.42±0.23	76.83±0.18
AMFM	**46.42±0.46**	**50.60±0.14**	**61.30±0.48**	**70.92±0.18**	**78.09±0.24**
AMFM+WAPCM	-	**54.72±0.41**	**61.95±0.38**	**71.49±0.21**	**78.66±0.16**

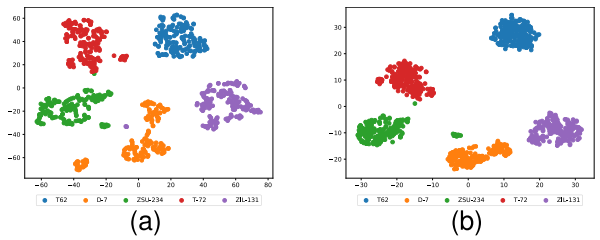


Fig. 11. Visualization results of embedding vectors distribution on the MSTAR dataset using t-SNE. (a) AMFM. (b) I2C metric.

by utilizing AMFM, we can effectively integrate the local metric module (I2C) and the global metric module (BDC) to obtain discriminative feature representations, thereby further improving the accuracy of target classification.

2) Effectiveness of WAPCM: In this section, we will validate the effectiveness of WAPCM through a series of experiments on three datasets MSTAR (five-way), OpenSARship (three-way), and SAR-AIRcraft (three-way). As shown in Tables V–VII, the classification accuracy improves when using WAPCM under different shot settings. We use the t-SNE algorithm to visualize the prototypes in two different scenarios on the MSTAR and OpenSARship datasets, as shown in Fig. 12. It is clear that the prototypes computed using the two different methods are distinct. We can observe that some samples deviate from the cluster centers, which may be due to noise or outliers in the samples. Therefore, directly averaging the samples within each class to compute the prototype is unreliable, as this method does not consider the weight differences among the samples and may overlook important

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:45:04 UTC from IEEE Xplore. Restrictions apply.

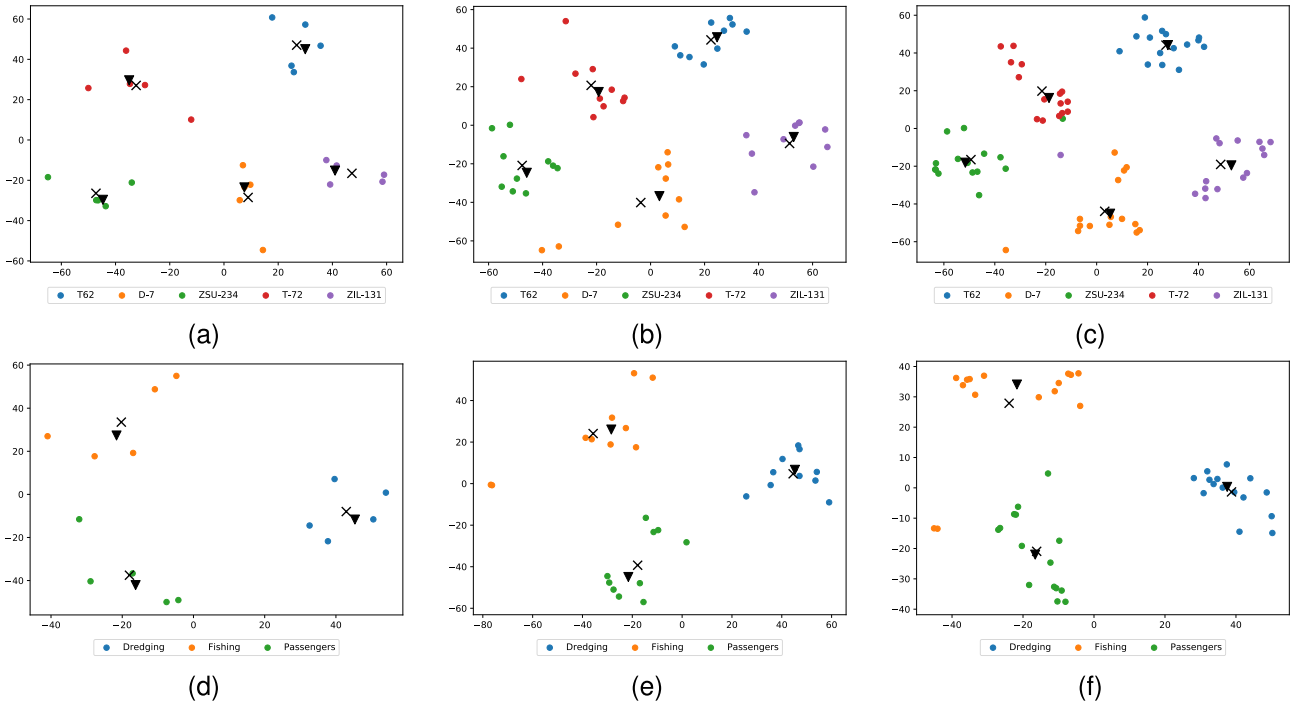


Fig. 12. "×" and "∇," respectively, represent the comparison of two different prototype calculation methods under different settings displayed on the MSTAR (a)–(c) and OpenSARship (d)–(f) datasets. The symbol "×" represents the prototypes obtained directly by averaging, while the symbol "∇" denotes the prototypes obtained through WAPCM. (a) five-way five-shot. (b) five-way ten-shot. (c) five-way 15-shot. (d) three-way five-shot. (e) three-way ten-shot. (f) three-way 15-shot.

TABLE VIII

CLASSIFICATION ACCURACY OF OUR GLMNET UNDER DIFFERENT VALUES OF $k \in \{1, 3, 5, 7\}$ ON DATASETS MSTAR (FIVE-WAY 15-SHOT), OPENSARSHIP (THREE-WAY 15-SHOT), AND SAR-AIRCRAFT (THREE-WAY 15-SHOT), RESPECTIVELY

Dataset	Accuracy(%)			
	k=1	k=3	k=5	k=7
MSTAR	84.93±0.23	**86.28±0.12**	85.85±0.14	85.26±0.21
OpenSARship	86.61±0.19	**87.74±0.18**	86.89±0.21	86.18±0.18
SAR-AIRcraft	76.85±0.25	**78.66±0.16**	78.24±0.19	77.29±0.22

information, further affecting the accuracy of the classification performance. Our designed WAPCM takes into account the differences in sample weights and can calculate more reliable prototypes, thereby improving the accuracy of target classification.

3) Effect of the Number k of Neighbors: In the I2C metric module, we need to find the k nearest neighbors in the support classes for each local descriptor of the query image. Next, we measure the similarity between the query image and a specific class based on the I2C metric. Therefore, selecting the appropriate hyperparameter k is crucial. To determine the optimal value of k, we conducted five-way 15-shot tasks on the MSTAR dataset and three-way 15-shot tasks on the other two datasets (OpenSARship and SAR-AIRcraft). We experiment with different values of k ($k \in \{1, 3, 5, 7\}$), and the results are shown in Table VIII. From the experimental results, we observed that the model achieved the highest accuracy when k is set to 3. Therefore, in our model, we choose to set the value of k as 3 to achieve the best classification performance.

D. Contrast Experiments

1) Experiments Analysis: Based on the same feature extraction network, we compare the average test accuracies of GLMnet and other classic few-shot classification models (such as Protonet [33], Baseline++ [55], DN4 [36], Meta-baseline [56], DMN4 [57], and CPN [58]) on three SAR datasets. The specific results are shown in Tables IX–XI. From the results, it is evident that our proposed GLMnet achieved the best classification accuracy across different shot settings. Particularly, our model improved performance by 2.53% and 2.92% in the five-way five-shot and three-way two-shot settings, respectively. Furthermore, the proposed algorithm can improve 6.27% on the SAR-AIRcraft dataset under the three-way two-shot setting. The improvement in accuracy is not significant in the five-way one-shot (MSTAR dataset) and three-way one-shot (OpenSARship and SAR-AIRcraft datasets) settings. This is mainly because, in these settings, there is only one support class sample, which cannot be utilized by our proposed WAPCM. Based on the aforementioned experimental analysis, our GLMnet can obtain more

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:45:04 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
ACCURACY COMPARISONS ON THE MSTAR DATASET UNDER FIVE-WAY K-SHOT SETTINGS ($K \in [1, 2, 5, 10, 15]$)

Method	Accuracy(%)				
	5-way 1-shot	5-way 2-shot	5-way 5-shot	5-way 10-shot	5-way 15-shot
Protonet	51.53±0.38	54.30±0.33	55.55±0.26	72.75±0.23	74.16±0.23
Baseline++	50.59±0.35	56.74±0.30	64.75±0.27	73.19±0.24	76.63±0.24
DN4	52.32±0.46	57.03±0.37	68.68±0.23	79.28±0.12	84.79±0.21
Meta-baseline	52.14±0.34	55.01±0.33	58.83±0.28	80.03±0.21	65.59±0.25
CPN	49.22±0.53	53.70±0.50	54.54±0.38	65.96±0.39	72.63±0.21
DMN4	52.38±0.45	57.42±0.41	70.64±0.26	80.12±0.18	85.28±0.22
GLMnet(ours)	**52.83±0.33**	**58.64±0.27**	**73.17±0.16**	**80.68±0.24**	**86.28±0.12**

TABLE X
ACCURACY COMPARISONS ON THE OPENSARSHIP DATASET UNDER THREE-WAY K-SHOT SETTINGS ($K \in [1, 2, 5, 10, 15]$)

Method	Accuracy(%)				
	3-way 1-shot	3-way 2-shot	3-way 5-shot	3-way 10-shot	3-way 15-shot
Protonet	70.81±0.45	73.49±0.33	79.17±0.26	85.36±0.21	82.01±0.23
Baseline++	67.83±0.45	73.17±0.34	79.06±0.28	83.14±0.24	84.17±0.23
DN4	70.28±0.25	72.19±0.21	79.81±0.23	85.26±0.37	86.19±0.22
Meta-baseline	66.06±0.54	70.50±0.43	76.55±0.40	79.22±0.23	79.32±0.22
CPN	70.56±0.23	73.70±0.20	76.43±0.21	79.30±0.38	80.93±0.18
DMN4	65.22±0.42	71.46±0.43	81.30±0.34	76.59±0.27	79.67±0.21
GLMnet(ours)	**71.35±0.32**	**76.62±0.33**	**81.94±0.23**	**86.21±0.17**	**87.74±0.18**

TABLE XI
ACCURACY COMPARISONS ON THE SAR-AIRCRAFT DATASET UNDER THREE-WAY K-SHOT SETTINGS ($K \in [1, 2, 5, 10, 15]$)

Method	Accuracy(%)				
	3-way 1-shot	3-way 2-shot	3-way 5-shot	3-way 10-shot	3-way 15-shot
Protonet	45.41±0.48	48.75±0.42	54.36±0.40	61.98±0.35	63.44±0.28
Baseline++	44.13±0.32	51.15±0.43	59.84±0.31	66.90±0.18	70.79±0.20
DN4	40.05±0.60	49.97±0.22	60.74±0.28	70.42±0.23	76.83±0.18
Meta-baseline	42.87±0.45	48.87±0.36	54.61±0.26	59.89±0.32	59.96±0.21
CPN	45.61±0.27	50.17±0.36	51.17±0.16	58.16±0.42	70.93±0.12
DMN4	45.56±0.39	48.06±0.42	59.48±0.31	70.41±0.25	77.43±0.18
GLMnet(ours)	**46.42±0.46**	**57.42±0.41**	**61.95±0.38**	**71.49±0.21**	**78.66±0.16**

reliable support class prototypes and achieve discriminative metric performance, thereby enhancing the accuracy of target classification.

2) Explanation of Lower Accuracy With Large-Shot Than Small-Shot: We have observed that, during the testing on the MSTAR dataset, Meta-baseline achieves higher classification accuracy in the ten-shot setting compared to the 15-shot setting. Similarly, for the OpenSARship dataset, Protonet exhibits higher classification accuracy in the ten-shot setting compared to the 15-shot setting. When the shot increases, the decrease in accuracy can be attributed to two main factors: data noise and differences between categories.

First, as the shot increases, the number of samples in the training set also increases. However, the additional training samples may contain more noise and inconsistencies, making it difficult for the model to accurately learn the true features of the categories. This noise can come from labeling errors, image quality issues, or variations within categories. Therefore, increasing the shot may introduce more interference, causing the model to be more disturbed during the training phase and affecting its performance during the testing phase.

Second, as the shot increases, the differences between categories become more pronounced. When the shot is small, the differences between categories may be relatively small, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:45:04 UTC from IEEE Xplore. Restrictions apply.

the model can more easily learn the distinguishing features between them. However, as the shot increases and the number of samples per category increases, more intraclass variations and interclass differences may be introduced. This makes it more challenging for the model to determine decision boundaries, as the boundaries between categories become blurred and intraclass variations become more complex. Therefore, in higher shot settings, the model may encounter more difficulties in accurately distinguishing between different categories.

In conclusion, increasing the shot may introduce more noise and interference, while also increasing the differences between categories. These factors collectively lead to a decrease in performance for the model in higher shot settings. To address this issue, it may be necessary to perform more refined data cleaning and preprocessing, as well as adopt more complex models or improved algorithms to handle the increased complexity of intercategory differences and noise issues. Therefore, in practical applications, the choice of an appropriate K value needs to be carefully considered depending on the specific circumstances.

VI. Conclusion

This article introduces a discriminative metric network called GLMnet for few-shot SAR target classification. Due to the limitation of local descriptors in capturing only local information of the image while ignoring global relationships, the features of the image cannot be fully represented. Therefore, we introduce a BDC metric module to capture the global relationships between descriptors. Furthermore, we design an AMFM to integrate these two metric relationships and thus achieve more discriminative metric performance. Additionally, during the BDC metric process, we also devise a WAPCM to address the issue of prototypes deviating from the true distribution due to outliers or noisy samples, thereby obtaining more reliable prototypes. Experimental results on three SAR datasets illustrate that GLMnet outperforms other metric-based FSL algorithms in terms of classification performance.

References

[1] A. Theron, J. Engelbrecht, J. Kemp, W. Kleynhans, and T. Turnbull, "Detection of sinkhole precursors through SAR interferometry: Radar and geological considerations," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 6, pp. 871–875, Jun. 2017.

[2] G. D. Martino, A. Iodice, D. Riccio, and G. Ruello, "A novel approach for disaster monitoring: Fractal models and tools," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 6, pp. 1559–1570, Jun. 2007.

[3] S. Wollstadt, P. López-Dekker, F. De Zan, and M. Younis, "Design principles and considerations for spaceborne ATI SAR-based observations of ocean surface velocity vectors," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4500–4519, Aug. 2017.

[4] B. Ding, G. Wen, C. Ma, and X. Yang, "Decision fusion based on physically relevant features for SAR ATR," *IET Radar, Sonar Navigat.*, vol. 11, no. 4, pp. 682–690, Apr. 2017.

[5] Y. Huang, G. Liao, Z. Zhang, Y. Xiang, J. Li, and A. Nehorai, "SAR automatic target recognition using joint low-rank and sparse multiview denoising," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 10, pp. 1570–1574, Oct. 2018.

[6] L. Tao, X. Jiang, X. Liu, Z. Li, and Z. Zhou, "Multiscale supervised kernel dictionary learning for SAR target recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 9, pp. 6281–6297, Sep. 2020.

[7] C. Wang et al., "Label noise modeling and correction via loss curve fitting for SAR ATR," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5216210.

[8] Z. Zhou, Z. Cao, and Y. Pi, "Subdictionary-based joint sparse representation for SAR target recognition using multilevel reconstruction," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6877–6887, 2019.

[9] Y. LeCun et al., "Backpropagation applied to handwritten zip code recognition," *Neural Comput.*, vol. 1, no. 4, pp. 541–551, Dec. 1989.

[10] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 60, no. 6, 2017, pp. 84–90.

[11] Z. Huang, Z. Pan, and B. Lei, "Transfer learning with deep convolutional neural network for SAR target classification with limited labeled data," *Remote Sens.*, vol. 9, no. 9, p. 907, Aug. 2017.

[12] S. Chen, H. Wang, F. Xu, and Y.-Q. Jin, "Target classification using the deep convolutional networks for SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4806–4817, Aug. 2016.

[13] J. Ai, Y. Mao, Q. Luo, L. Jia, and M. Xing, "SAR target classification using the multikernel-size feature fusion-based convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5214313.

[14] R. Xue, X. Bai, and F. Zhou, "Spatial–temporal ensemble convolution for sequence SAR target classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1250–1262, Feb. 2020.

[15] L. Yu, Y. Hu, X. Xie, Y. Lin, and W. Hong, "Complex-valued full convolutional neural network for SAR target classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 10, pp. 1752–1756, Oct. 2020.

[16] F. Zhang, Y. Wang, J. Ni, Y. Zhou, and W. Hu, "SAR target small sample recognition based on CNN cascaded features and AdaBoost rotation forest," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 6, pp. 1008–1012, Jun. 2020.

[17] J. Liu, M. Xing, H. Yu, and G. Sun, "EFTL: Complex convolutional networks with electromagnetic feature transfer learning for SAR target recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2021.

[18] B. Peng, B. Peng, J. Zhou, J. Xia, and L. Liu, "Speckle-variant attack: Toward transferable adversarial attack to SAR target recognition," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[19] S. Feng, K. Ji, F. Wang, L. Zhang, X. Ma, and G. Kuang, "PAN: Part attention network integrating electromagnetic characteristics for interpretable SAR vehicle target recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5204617.

[20] Y. Zhang, X. Guo, H. Leung, and L. Li, "Cross-task and cross-domain SAR target recognition: A meta-transfer learning approach," *Pattern Recognit.*, vol. 138, Jun. 2023, Art. no. 109402.

[21] Z. Wang, L. Du, J. Mao, B. Liu, and D. Yang, "SAR target detection based on SSD with data augmentation and transfer learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 1, pp. 150–154, Jan. 2019.

[22] S. Du, J. Hong, Y. Wang, K. Xing, and T. Qiu, "Physical-related feature extraction from simulated SAR image based on the adversarial encoding network for data augmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[23] X. Hao, L. Liu, R. Yang, L. Yin, L. Zhang, and X. Li, "A review of data augmentation methods of remote sensing image target recognition," *Remote Sens.*, vol. 15, no. 3, p. 827, Feb. 2023.

[24] Y. Sun, Y. Wang, H. Liu, N. Wang, and J. Wang, "SAR target recognition with limited training data based on angular rotation generative network," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 11, pp. 1928–1932, Nov. 2020.

[25] Z. Ren, B. Hou, Q. Wu, Z. Wen, and L. Jiao, "A distribution and structure match generative adversarial network for SAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3864–3880, Jun. 2020.

[26] C. Du and L. Zhang, "Adversarial attack for SAR target recognition based on UNet-generative adversarial network," *Remote Sens.*, vol. 13, no. 21, p. 4358, Oct. 2021.

[27] C. Zhang, Y. Wang, H. Liu, Y. Sun, and L. Hu, "SAR target recognition using only simulated data for training by hierarchically combining CNN and image similarity," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[28] X. Zhang, Y. Luo, and L. Hu, "Semi-supervised SAR ATR via Epoch- and uncertainty-aware pseudo-label exploitation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5209015.

[29] Z. Sun, J. Li, P. Liu, W. Cao, T. Yu, and X. Gu, "SAR image classification using greedy hierarchical learning with unsupervised stacked CAEs," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5721–5739, Jul. 2021.

[30] L. Fe-Fei et al., "A Bayesian approach to unsupervised one-shot learning of object categories," in *Proc. 9th IEEE Int. Conf. Comput. Vis.*, Oct. 2003, pp. 1134–1141.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:45:04 UTC from IEEE Xplore. Restrictions apply.

[31] T. Hospedales, A. Antoniou, P. Micaelli, and A. Storkey, "Meta-learning in neural networks: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 9, pp. 5149–5169, Sep. 2022.

[32] G. Koch et al., "Siamese neural networks for one-shot image recognition," in *Proc. Int. Conf. Mach. Learn.*, Lille, 2015, vol. 2, no. 1, pp. 1–8.

[33] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for few-shot learning," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 4080–4090.

[34] V. Garcia and J. Bruna, "Few-shot learning with graph neural networks," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2018, pp. 1–13.

[35] F. Sung et al., "Learning to compare: Relation network for few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1199–1208.

[36] W. Li, L. Wang, J. Xu, J. Huo, Y. Gao, and J. Luo, "Revisiting local descriptor based image-to-class measure for few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 7260–7268.

[37] D. Wertheimer and B. Hariharan, "Few-shot learning with localization in realistic settings," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6551–6560.

[38] W. Li, L. Wang, J. Huo, Y. Shi, Y. Gao, and J. Luo, "Asymmetric distribution measure for few-shot learning," in *Proc. 29th Int. Joint Conf. Artif. Intell.*, 2020, pp. 1–7.

[39] C. Zhang, Y. Cai, G. Lin, and C. Shen, "DeepEMD: Few-shot image classification with differentiable earth mover's distance and structured classifiers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 12203–12213.

[40] J. Xie, F. Long, J. Lv, Q. Wang, and P. Li, "Joint distribution matters: Deep Brownian distance covariance for few-shot classification," in *Proc. CVPR*, 2022, pp. 7972–7981.

[41] L. Wang, X. Bai, C. Gong, and F. Zhou, "Hybrid inference network for few-shot SAR automatic target recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9257–9269, Nov. 2021.

[42] M. Yang, X. Bai, L. Wang, and F. Zhou, "Mixed loss graph attention network for few-shot SAR target classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5216613.

[43] H. Ren et al., "Transductive prototypical attention reasoning network for few-shot SAR target recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5206813.

[44] L. Zhang et al., "Domain knowledge powered two-stream deep network for few-shot SAR vehicle recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5215315.

[45] J. Cai, Y. Zhang, J. Guo, X. Zhao, J. Lv, and Y. Hu, "ST-PN: A spatial transformed prototypical network for few-shot SAR image classification," *Remote Sens.*, vol. 14, no. 9, p. 2019, Apr. 2022.

[46] L. Wang, X. Yang, H. Tan, X. Bai, and F. Zhou, "Few-shot class-incremental SAR target recognition based on hierarchical embedding and incremental evolutionary network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5204111.

[47] O. Vinyals et al., "Matching networks for one shot learning," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 3630–3638.

[48] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 448–456.

[49] A. L. Maas et al., "Rectifier nonlinearities improve neural network acoustic models," in *Proc. Int. Conf. Mach. Learn.*, 2013, vol. 30, no. 1, pp. 1–3.

[50] X. Glorot, A. Bordes, and Y. Bengio, "Deep sparse rectifier neural networks," in *Proc. 14th Int. Conf. Artif. Intell. Statist.*, 2011, pp. 315–323.

[51] A. Vaswani et al., "Attention is all you need," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, vol. 30, pp. 1–11, 2017.

[52] L. Huang et al., "OpenSARShip: A dataset dedicated to Sentinel-1 ship interpretation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 1, pp. 195–208, Jan. 2018.

[53] S. Xian, W. Zhirui, S. Yuanrui, D. Wenhui, Z. Yue, and F. Kun, "AIR-SARShip-1.0: High-resolution SAR ship detection dataset," *Radars*, vol. 8, no. 6, pp. 852–863, 2019.

[54] W. Zhirui, K. Yuzhuo, Z. Xuan, W. Yuelei, Z. Ting, and S. Xian, "SAR-aircraft-1.0: High-resolution SAR aircraft detection," *J. Radars*, vol. 12, no. 4, pp. 1–17, 2023.

[55] W.-Y. Chen, Y.-C. Liu, Z. Kira, Y.-C. F. Wang, and J.-B. Huang, "A closer look at few-shot classification," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019, pp. 1–16.

[56] Y. Chen, Z. Liu, H. Xu, T. Darrell, and X. Wang, "Meta-baseline: Exploring simple meta-learning for few-shot learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9062–9071.

[57] Y. Liu, T. Zheng, J. Song, D. Cai, and X. He, "DMN4: Few-shot learning via discriminative mutual nearest neighbor neural network," in *Proc. AAAI Conf. Artif. Intell.*, vol. 36, no. 2, pp. 1828–1836, Jun. 2022.

[58] Q. Lyu and W. Wang, "Compositional prototypical networks for few-shot classification," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 1–9.



Jia Zheng received the bachelor's degree in electronic information engineering from Xinyang Normal University, Xinyang, China, in 2015, and the master's degree in control science and engineering from Dalian University, Dalian, China, in 2018. He is currently pursuing the Ph.D. degree in signal and information processing with the National Key Laboratory of Radar Signal Processing, Xidian University, Xi'an, China.

His main research interests include radar automatic target recognition, few-shot learning, and deep learning.



Ming Li (Member, IEEE) received the B.S. degree in electrical engineering and the M.S. and Ph.D. degrees in signal processing from Xidian University, Xi'an, China, in 1987, 1990, and 2007, respectively.

In 1987, he joined the Department of Electronic Engineering, Xidian University, where he is currently a Professor with the National Key Laboratory of Radar Signal Processing. His research interests include adaptive signal processing, detection theory, ultrawideband, and synthetic aperture radar image processing.



Xiang Li received the B.S. and M.S. degrees from the National Laboratory of Radar Signal Processing, Xidian University, Xi'an, China, in 2009 and 2012, respectively. He is currently pursuing the Ph.D. degree with the Department of Electronic Engineering, Tsinghua University, Beijing, China.

He is a Researcher with the Beijing Institute of Radio Measurement. His research interests include radar system design, cognitive SAR/ISAR imaging, and moving target parameter estimation.



Peng Zhang (Member, IEEE) received the B.S. degree in electronic and information engineering and the M.S. and Ph.D. degrees in signal and information processing from Xidian University, Xi'an, China, in 2006, 2009, and 2012, respectively.

He is currently an Associate Professor with the National Key Laboratory of Radar Signal Processing, Xidian University. His main research interests include SAR image interpretation and statistical learning theory.



Yan Wu (Member, IEEE) received the B.S. degree in information processing and the M.S. and Ph.D. degrees in signal and information processing from Xidian University, Xi'an, China, in 1987, 1998, and 2003, respectively.

From 2003 to 2005, she was a Post-Doctoral Fellow with the National Key Laboratory of Radar Signal Processing, Xi'an. Since 2005, she has been a Professor with the Department of Electronic Engineering, Xidian University. Her research interests include remote sensing image analysis and interpretation, data fusion of multisensor images, synthetic aperture radar autotarget recognition, and statistical learning theory and application.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:45:04 UTC from IEEE Xplore. Restrictions apply.